Michelle Gurevich

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Research interests

I am interested in using gravitational waves to study black hole formation and the cosmological constant, and to develop modified theories of gravity.

Education

2020 – Present **MSc in Physics with Extended Research** – Imperial College London
Modified dispersion relations for gravitational waves. Supervisor: Carlo Contaldi.
Machine learning for 21cm studies and cosmology. Supervisor: Jonathan Pritchard.
Chaos and fractals. completed for Research Skills in MATLAB.

2014 – 2018 **BSc in Mathematics** – University of Michigan, Ann Arbor
Grade achieved: 1st class. Supervisor: Stephen DeBacker.

Awards, fellowships, and honors

2021 4Cs Science Communication Competition
Selected to present research to a panel of expert judges in the Graduate School of Imperial College London. One of a group of 18 postgraduates picked from applicant pool.

Publications

2019 **Search for RR Lyrae stars in DES ultra-faint systems: Grus I, Kim 2, Phoenix II, and Grus II**.
Clara Eugenia Martinez Vazquez, Kathy Vivas, Michelle Gurevich, et al.
Monthly Notices of the Royal Astronomical Society, Volume 490, Issue 2.

Research experience

2021 – 2022 **Imperial College London** – London, UK
Supervisor: Carlo Contaldi.

- Extended research project focusing in the development of a phenomenological model of modified waveforms.
- Recreated PyCBC frequency and time domain waveforms (assuming a stationary phase approximation) to develop a phenomenological model for modifying gravitational wave signals in LIGO and LISA frequency bands.
- Performed a Fisher analysis to identify parameter behavior and constrain data.
- Presented poster to department.

Summer 2016 **Cerro Tololo Inter-American Observatory** – La Serena, Chile
Supervisor: Kathy Vivas.

- Collaborated on a research project using RR Lyrae variable stars as standard candles to measure galactic distances.
- Took observations at the Cerro Tololo 4-m telescope, reduced CCD images, organized and interpreted data catalogues, wrote IRAF scripts to optimize star detection, and presented results.
- Scripted various command line processes and implemented a mask multiplier, reducing total galaxy reduction time by a factor of 16.

Summer 2017 **German Tech Entrepreneurship Center** – Berlin, Germany

- Helped implement a signal processing algorithm utilizing empirical mode decomposition and discrete wavelet transform for EEG data reconciliation.
- Researched sorting and ranking algorithms to develop internal search engine.
- Drafted tech analyses of interviews with start-up founders.

Industry experience

June 2018 – September 2020

JPMorgan Chase & Co. (Software Engineer II) – New York, NY

- Wrote machine learning algorithm for reducing false positive flagging of SWIFT payment messages, set up and managed database of 30,000 processed messages, and used results to predict validity of critical message fields.
- Developed APIs for validating SWIFT payment messages as well as a parsing and-validation protocal for IBANs, wrote scripts for generating and anonymizing sensitivedata.
- Designed UI using React for two platform applications and incorporated exception handling for production and testing environments.
- Set up cross-currency shocks for emerging markets and aggregated scenarios for instruments with exposure to multiple risk factors; configured batch process jobs for sensitivity stress calculations.
- Troubleshot code issues for various trading groups during production CCAR runs to present solvency reports to the federal government.
- Scripted mark-to-market report for 2 environments and over 50 trading groups to reconcile inconsistencies between pricing done by front and back office.

Talks and tutorials

February 2022

Modified dispersion relations for gravitational waves
PLANCKS 2022 UK and Ireland

Mentorship and outreach

2018 - 2019

Force for Good, AmeriCorps

- Developed database for Green City Force (GCF), a NYC-based AmeriCorps non-profit, as part of JP Morgan volunteering network.
- Coordinated with GCF to digitize their files using a database. When we started, GCF used pencil and paper for most of their documentation, including sensitive client data. By introducing a secure system, we ensured this personal data would be stored in a safe, centralized, easily-accessible way.
- Helped build an app to make signing on new participants easier for GCF volunteers while safe-keeping data in an encrypted layer.

2015 - 2018 **Math Club**

- Taught middle school students from surrounding schools the basics of mathematical proof writing and logic puzzles.
- Helped coordinate fee waivers to allow for increased participation from underrepresented groups.

2016 - 2018 **Student Astronomical Society**

- Completed training to use the 0.4-m telescope located in the observatory of Angell Hall at the University of Michigan.
- Operated the telescope and helped run public open houses at the planetarium as part of the Student Astronomical Society (SAS).
- Served as treasurer of SAS, which gave me a unique opportunity to coordinate with members of the community and increase access to events hosted by the astronomy department.

Professional memberships

Since 2022 Institute of Physics (IoP)
Member

Since 2020 American Astronomical Society (AAS)
Graduate student member

Technical skills

Programming languages
Proficient in: Python, C++, Java, MATLAB
Familiar with: Mathematics, IRAF, React

Software
LaTeX, Git

Languages
English, Spanish, Russian

Telescope time
14 – 17 July, 2016 at the Blanco 4-m telescope in Chile